

Siraj TM

BSMS Data Science, 5TH year
IISER TVM

CONTACT

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EXPERIENCE

MAJOR THESIS

Guide - Dr. Raji Susan Mathew
August 2025 - Present

Developed an interpretable voxel-wise deep reinforcement learning framework for Quantitative Susceptibility Mapping (QSM) Reconstruction.

INTERNSHIP

Guide - Dr. Raji Susan Mathew
In Collaboration with Dr. Navchetan Awasthi, University of Amsterdam
December 2024 - August 2025

Worked on improving generalizability in QSM reconstruction, by incorporating contextual input-target pairs. The resulting work has been accepted at ICVGIP'25.

INTERNSHIP

Guide - Dr. Raji Susan Mathew and Dr. Reji
November 2024 - July 2025

Worked on remote sensing change captioning using state space models by incorporating change detection using graph neural networks. The resulting work has been accepted at MIGARS 2026

Internship

Guide - Dr. Chandrakala Meena
Collaborated with Dr. Sandeep Chowdhary, IIASA, Laxenburg, Austria
August 2023 - July 2025

Analysed faculty hiring networks between India and the rest of the world with the aim of measuring the impact of different academic migration trajectories on careers

PUBLICATIONS

Accepted:

T. Apotheker, **Siraj T. M**, R. S. Mathew, and N. Awasthi. "Context-qsmnet: Enhancing Generalization in Quantitative Susceptibility Mapping with Context-Aware Deep Neural Networks." **Accepted at the Indian Conference on Computer Vision, Graphics, and Image Processing (ICVGIP), 2025.**

Siraj T. M, Reji V., R. S. Mathew. "Graph-Guided Change Detection and State Space Model Captioning with Semantic Uncertainty Estimation " **Accepted at MIGARS 2026.**

Under Review:

Siraj T. M, Meenkashi S., Reji V., R. S. Mathew. RL-QSM: An Interpretable Reinforcement Learning-Based Reconstruction for Quantitative Susceptibility Mapping. **Submitted to MICCAI 2026.**

Siraj T. M., Hari Krishnan S, Mathew Vincent, Sandeep Chowdhary and Chandrakala Meena. "Scientific mobility patterns of Indian researchers: Impact on career growth." **Submitted to Quantitative Science Studies (QSS)**, 2025. Available on arxiv

EDUCATION

INDIAN INSTITUTE OF SCIENCE, EDUCATION AND RESEARCH,
THRIUVANANTHAPURAM
BSMS, I2 Data Science
2021 - Present
CGPA: 8.00/10
Major CGPA: 8.4/10
TOEFL : 112/120

Bhavans Varuna Vidhyalaya, Thrikkakkara, Kochi
AISSE (2019) and AISSCE (2021) - **CBSE AISSCE: 96.8 %**

SKILLS

- QSM / MRI Reconstruction, Medical Imaging, Computer Vision
 - Python, C, MATLAB, C++, R, HTML, CSS, DBMS, JavaScript
 - Pytorch, Tensorflow, OpenCV, CUDA, MPI, OpenMP
 - Git, Bash, Linux,
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AWARDS

INSPIRE Scholarship awarded by DST-GOI Awarded to the top 1% of students based on Class 12 board exam performance. Scholarship for pursuing Natural and Basic Sciences at the BSMS level.

Conferences & Symposiums

- **ISMRM India Chapter 2026** - e-Poster (received certificate of merit)
 - **Frontier Symposium In Data Science (FS-DS) IISER TVM 2026** - Oral Presentation First runner up.
 - **ICVGIP 2025** - Poster Presentation
 - **FS-DS IISER TVM 2025** - Oral Presentation
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COURSES

AI and Medical Imaging | Artificial Intelligence | Machine Learning I & II |
Data Structures | Mathematical Statistics | Parallel and Distributed
Computing | High Performance Computing | Linear Algebra | Optimization

REFERENCES

Dr. Raji Susan Mathew
Master's Thesis Guide
Assistant Professor, Department of Data Science
IISER Thiruvananthapuram, Kerala, India
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Dr. Navchetan Awasthi
Internship Guide
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Intelligence and Data Science
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Dr. Alwin Poulose
Course Instructor
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